

SENTINEL: Evaluating Pipeline Robustness to Distributional Shifts

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Abstract—Ensuring the robustness of data preprocessing pipelines is essential for maintaining the reliability of machine learning model performance in the face of real-world data shifts. Traditional methods optimize preprocessing sequences for specific datasets but often overlook their vulnerability to future data variations. We demonstrate SENTINEL, a system for evaluating the robustness of data preprocessing pipelines under real-world covariate shift. SENTINEL quantifies component-wise susceptibility using a composite vulnerability score (VS) and learns a lightweight linear mapping from VS to downstream performance change. For any given pipeline and drift scenario, SENTINEL produces an explanation that classifies the pipeline as vulnerable or non-vulnerable and indicates the likely contribution to performance degradation. For the shift-vulnerable scenarios, we further suggest repaired pipelines for system maintainers that preserve the model performance without re-optimization. The proposed framework delivers a risk-aware assessment, empowering practitioners to anticipate potential performance changes and adapt their pipeline repair strategies accordingly. On multiple real datasets, SENTINEL delivers a risk-aware, practitioner-facing report that pinpoints brittle stages and recommends repairs to mitigate drift-induced failures prior to deployment.

I. INTRODUCTION

Machine learning (ML) pipelines are now central to data-driven decision-making across domains. These end-to-end workflows compose a sequence of preprocessing operations (e.g., missing-value imputation, outlier detection, encoding) that transform raw inputs into model-ready features. Optimizing such pipelines is an essential task in the ML lifecycle: conventional methods search for the sequence of components that maximizes performance on a given training set [1], and modern AutoML systems [2] automate selection and tuning pipeline configurations to optimize model performance provided the deployment data are drawn from the same distribution as the training data. In practice, however, deployed pipelines face evolving data. Incoming streams frequently deviate from the training distribution, a phenomenon broadly referred to as **data shift** [3]. Two primary forms of data shift are covariate shift—changes in the feature distribution while the conditional label mechanism remains stable [4]—and concept shift, where the relationship between inputs and outputs itself changes over time [5]. Such shifts can substantially degrade model performance if left unaddressed. In this demonstration, we focus on covariate shift, as it is a significant form of data

shift [4] and its effects can be characterized by changes in first- and higher-order statistics of the data [6].

We illustrate the challenges faced in this scenario through the following example.

Example 1. Consider a classification task with *AdultIncome* [7] dataset to predict whether an individual’s income exceeds \$50K given tabular demographics. An AutoML [2] procedure optimized the preprocessing pipeline on the training distribution as (imputation:mean $\rightarrow$ scaling:standard $\rightarrow$ outlier handling:IQR) followed by a logistic regression classifier. Post-deployment, a drift in data distribution is observed: 10% missing values and 15% of outliers arise on the feature *Age* which now becomes heavier-tailed. While the classifier is unchanged, these shifts propagate through the preprocessing pipeline and degrade model performance by 6%. The core problem is to quantify pipeline susceptibility to such shifts, decide whether the end-to-end pipeline remains within a tolerable risk threshold, and—if not—identify model-agnostic pipeline repair (e.g., median vs. mean imputation, order swap) that preserve predictive utility on the drifted data.

To address these challenges, we demonstrate SENTINEL, a practitioner-facing system that answers a central question: *How vulnerable is an optimized preprocessing pipeline under distributional shift?* SENTINEL evaluates whether a pipeline optimized on the source distribution remains effective when the data distribution changes. It does so by computing a model-agnostic, component-wise vulnerability score (VS) that blends post-transform sensitivity with distributional dissimilarity, aggregating these signals to (i) localize vulnerable components, and (ii) attribute predicted performance loss. When pipeline vulnerability is detected—SENTINEL triggers pipeline repair that proposes reconfigurations (e.g., module substitutions and/or order adjustments) compliant with schema and constraints.

SENTINEL provides an interactive, end-to-end workflow that enables users to: (i) synthesize or import drift scenarios (single or composed; scoped to selected features or the full schema), (ii) inspect per-component VS with thresholded shift-vulnerable flags and an interpretable forecast of performance loss, (iii) set or calibrate the threshold τ to reflect operational risk tolerance and observe how classification (vulnerable vs. non-vulnerable), and (iv) receive and compare retraining-

Step 1. Select Dataset

AdultIncome
GermanCredit
ACSEmployment

Data Preview

	age	workclass	education	income
0	38	Private	Masters	<=50K
1	51	Private	Bachelors	<=50K
2	50	Gov	Masters	<=50K
3	40	Gov	Bachelors	<=50K
4	31	Self-emp	Bachelors	<=50K

Step 2. Set Pipeline & Classifier

Pipeline Step 1
Imputer
Simple Imputer (Mean)

Pipeline Step 2
Normalizer
Standard Scaler

Pipeline Step 3
Outlier Detector
IQR

Select Classifier
Logistic Regression
Logistic Regression
Random Forest
Support Vector Machine (SVM)

Selected Pipeline: Data → Simple Imputer (Mean) → Standard Scaler → IQR → Classification (Logistic Regression) → Predictions

Fig. 1: SENTINEL’s input module: User selects a dataset and a supported classifier, then composes a preprocessing pipeline by choosing components and specifying their execution order. SENTINEL validates the specification and flags invalid compositions.

free recommendations for alternative preprocessing sequences that satisfy a target performance goal. SENTINEL can be employed as a pre-deployment robustness assessment, or as an actionable mitigation tool at post-deployment as discussed in Example 1. All inputs (pipelines, models, drift scenarios) and outputs (scores, predictions, recommendations, and provenance such as seeds and scenario parameters) are exportable for reproducibility and side-by-side evaluation across datasets and back-ends. A detailed description of the system and its foundations appears in our prior work [8]. This demonstration makes the following contributions:

- We present SENTINEL, an end-to-end system that explains and evaluates ML pipeline robustness by quantifying pipeline vulnerability and a predictive mapping from vulnerability to change in the downstream performance—enabling risk-aware robustness assessment and pipeline adaptation without retraining the model.
- The demonstration allows users to simulate drifts, inspect per-component susceptibility with shift-vulnerable flags, and export a practitioner report one-click alongside actionable pipeline reconfiguration suggestions.
- We demonstrate SENTINEL on multiple real-world datasets under diverse drift settings, varying noise types and their configurations.

Most prior work detects data drift and then adapts models or costly end-to-end pipeline re-optimization, but rarely treats the preprocessing pipeline itself as the object of analysis. To our knowledge, SENTINEL is the first framework that quantifies pipeline vulnerability under drift and explains how these vulnerabilities propagate to the pipeline’s behavior.

II. SYSTEM FRAMEWORK

This section provides a brief overview of SENTINEL’s internal design; for a detailed description, we refer readers to our foundational work [8]. SENTINEL assesses and explains

the vulnerability of machine learning pipelines to distributional change by quantifying pipeline drift with complementary drift measures. It simulates post-deployment scenarios to evaluate robustness and produces a comprehensive practitioner report that highlights fragile pipeline status, estimates performance impact, and recommends pipeline repair for a specified task. In the following, we present a modular overview of SENTINEL.

Vulnerability Measurement. SENTINEL assigns each preprocessing stage P_k a *vulnerability score* (VS) that combines: (i) a post-transform *sensitivity* term comparing $P_k(\mathcal{D}_{\text{orig}})$ to $P_k(\mathcal{D}_{\text{shift}})$, and (ii) a *distributional dissimilarity* term (e.g., $D_{\text{KL}}(P_k(\mathcal{D}_{\text{orig}}) \parallel P_k(\mathcal{D}_{\text{shift}}))$) between the corresponding output distributions, weighted by (λ_1, λ_2) to balance local vs. global effects. Vulnerability scores are model-agnostic and computed per component, then aggregated to a pipeline-level that directly quantifies robustness risk attributable to drift.

Drift Simulation. Given a dataset, SENTINEL instantiates multiple, controlled covariate-drift scenarios (single or composed; feature-scoped or schema-wide). For each scenario, per-component scores are stacked into a *VS matrix* $[VS_{P_k}^{(j)}]_{k,j}$ that reveals components’ vulnerability across conditions. SENTINEL then analyzes VS trajectories as a function of perturbation intensity to calibrate a practical tolerable threshold τ ; components or pipelines whose scores exceed τ are flagged shift-vulnerable.

Performance Impact. Using the simulated scenarios and their observed performance changes, SENTINEL fits a lightweight regression $f : VS \rightarrow \Delta$ that predicts the degradation Δ directly from dataset-level vulnerability signals. Given a pipeline, this mapping enables fast, retraining-free risk estimation suitable for deployment checks. The resulting predictions, together with τ , drive downstream decisions (retain, monitor, or repair) and inform the ranked list of candidate pipeline adjustments.

Repair Search. When a given pipeline is flagged as vulner-

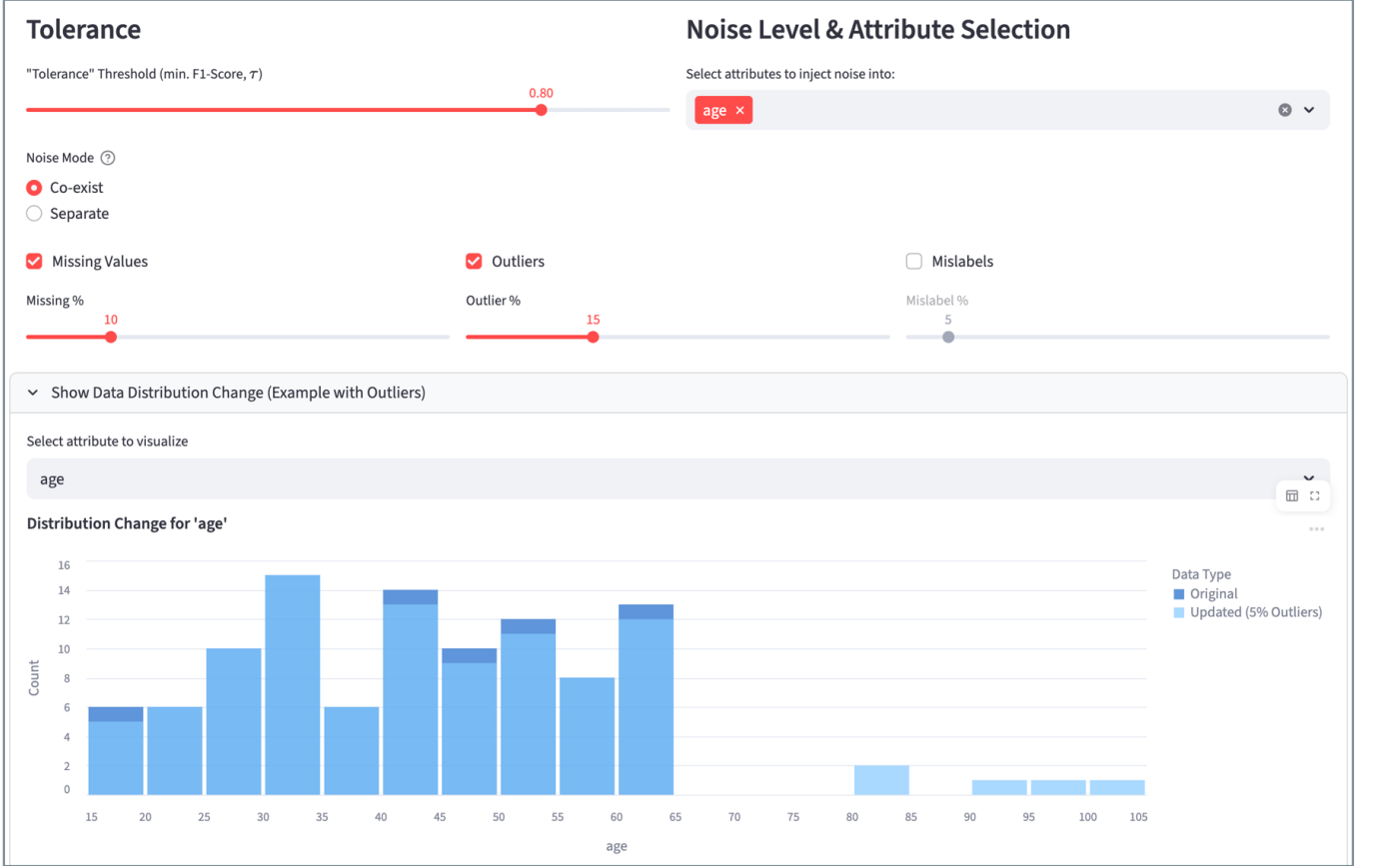


Fig. 2: Given the context, user configures a drift scenario, and SENTINEL synthetically generates the drifted dataset to evaluate.

able, SENTINEL invokes a pipeline repair phase that searches over feasible component choices (predefined in a global catalog) and orderings to minimize the pipeline-level vulnerability as a proxy for performance. Formally, let $\mathcal{S}$ denote the space of pipelines obtained by valid substitutions and permutations that complies defined constraints; SENTINEL solves

$$\min_{\mathcal{P}' \in \mathcal{S}} \text{VS}(\mathcal{P}') \quad \text{s.t.} \quad \text{VS}(\mathcal{P}') \leq \tau$$

and uses the learned mapping $f : \text{VS} \rightarrow \Delta$ to prioritize candidates by predicted degradation $\hat{\Delta}$. Following optimization, SENTINEL returns a top-k list of pipeline repair configurations.

SENTINEL consolidates these pieces into a practitioner-facing explanation that identifies vulnerable components, quantifies likely performance loss, and proposes actionable alternatives to preserve utility.

III. DEMONSTRATION OVERVIEW

A. Input Module

1) *Dataset and Classifier Selection:* We demonstrate SENTINEL on AdultIncome [7], a tabular dataset of demographic attributes for predicting whether an individual’s income exceeds \$50K. To broaden coverage, the demo also includes GermanCredit [9](Predicts loan approval/denial) and ACSEmployment [10](Predicts employed or not), enabling users to explore diverse application settings (Figure 1, Step 1). For modeling back-ends, we consider four representative classifiers: (i) logistic regression, (ii) random forest, (iii) support

vector machine, and (iv) a feed-forward neural network. Note that, while the measured vulnerability score (VS) is model-agnostic, the optimized preprocessing configurations and the resulting performance changes are model-dependent.

2) *Pipeline creation:* After selecting a dataset and classifier, the user composes a preprocessing pipeline from a component catalog (e.g., imputer, outlier detector; three components in the demo), specifies strategies (e.g., mean vs. median), and sets the execution order illustrated in Step 2 in Figure 1. SENTINEL validates the specification and flags invalid pipeline compositions or implicit component behaviors (e.g., an outlier detector that implicitly drops missing values if imputer not specified.). This resulting pipeline—whether sourced from a prior optimization run or defined ad hoc—serves as the unit of analysis, upon which SENTINEL computes vulnerability score.

3) *Simulating drift scenarios:* The user then configures a data drift scenario by selecting a drift type from types: missing value, outlier, perturbation, label flipping, along with a noise level, which controls the percentage of data to be perturbed (Figure 2). Depending on the deployment context, users may apply a single drift or compose multiple drifts concurrently, target specific feature subsets, or the entire schema. Upon execution, the system synthetically generates the shifted dataset and feeds it into the forward computations.

B. Output Module

1) *Robustness Report:* The first output is a vulnerability report at both the pipeline and component levels. Given a user-

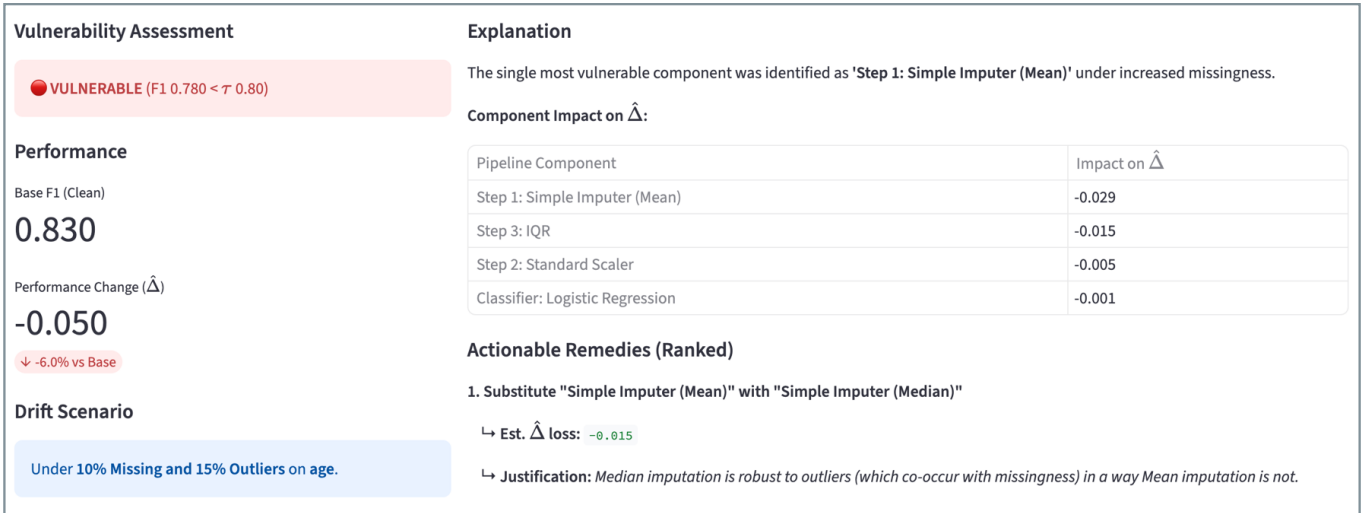


Fig. 3: A practitioner-facing pipeline evaluation report generated by SENTINEL, presenting the pipeline’s vulnerability score (VS) along with component-wise, and the estimated performance loss. Report also shows the ranked actionable pipeline repair.

specified tolerable threshold τ , SENTINEL classifies the current configuration as non-vulnerable or vulnerable (the aggregate pipeline score surpasses τ). For non-vulnerable pipelines, SENTINEL reports the per-component scores, the aggregate score, and a summary of predicted performance change $\hat{\Delta}$ and recommends retention of the existing configuration. For vulnerable pipelines, SENTINEL (i) identifies and ranks the most contributory component(s) by marginal impact on the pipeline-level score and predicted degradation, (ii) surfaces the drift scenarios under which vulnerability manifests (e.g., scope, intensity) to facilitate diagnosis, and (iii) issues actionable guidance for the subsequent reconfiguration phase.

2) *Explanation*: The second part of the output produces practitioner-facing explanations that connect measured vulnerability to operational decisions. SENTINEL synthesizes (i) a narrative summary of which stages are fragile and under what drift regimes; (ii) a causal-attribution style breakdown linking component scores to the predicted degradation $\hat{\Delta}$; and (iii) actionable top-K pipeline repairs that preserve utility without costly re-optimization. Remedies are presented as ranked candidates—module substitutions and/or order adjustments—that satisfy schema constraints and meet a user-specified performance budget. Each recommendation is accompanied by (a) the projected improvement in $\hat{\Delta}$ and reduction in VS, (b) provenance of the evidence used (drift scenario settings, seeds), and (c) a compact justification of repair.

IV. DEMONSTRATION SCENARIOS

On the AdultIncome dataset, following the scenario in Example 1, an optimized preprocessing pipeline (imputation: mean $\rightarrow$ scaling: standard $\rightarrow$ outlier handling: IQR) on the source distribution with a logistic regression classifier achieved 0.83 accuracy. Under a covariate-drift scenario in which missingness and outliers rose from 6% and 8% to 10% and 15% respectively, the ground-truth accuracy declined by $\approx 6\%$ (user-defined tolerable threshold is 3%). Within this context, SENTINEL correctly classifies the pipeline as

vulnerable (Figure 3), assigning a pipeline-level vulnerability score $VS = 7.8$ that exceeded the tolerable threshold of VS , $\tau = 4.2$, and forecasting a $\hat{\Delta} \approx 5\%$ performance loss (close to the observed 6%). The single most vulnerable component is identified as mean imputation under increased missingness. Following the repair ranking, changing the pipeline to—(imputation: median $\rightarrow$ scaling: standard $\rightarrow$ outlier handling: IQR)—reduces vulnerability to $VS = 2.4$ and yields an expected performance loss of $\hat{\Delta} \approx 1.5\%$ (observed $\approx 2\%$). In a follow-up experiment, evaluating the suggested repaired pipeline on the drifted dataset, SENTINEL classifies it as non-vulnerable.

AI-GENERATED CONTENT ACKNOWLEDGEMENT

The authors confirm that no AI-generated content was used in the preparation of this work. Grammar-checking tools were used only for language refinement.

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